

Bidirectional Copy-Paste for Semi-Supervised Medical Image Segmentation

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Abstract

In semi-supervised medical image segmentation, there exist empirical mismatch problems between labeled and unlabeled data distribution. The knowledge learned from the labeled data may be largely discarded if treating labeled and unlabeled data separately or in an inconsistent manner. We propose a straightforward method for alleviating the problem — copy-pasting labeled and unlabeled data bidirectionally, in a simple Mean Teacher architecture. The method encourages unlabeled data to learn comprehensive common semantics from the labeled data in both inward and outward directions. More importantly, the consistent learning procedure for labeled and unlabeled data can largely reduce the empirical distribution gap. In detail, we copy-paste a random crop from a labeled image (foreground) onto an unlabeled image (background) and an unlabeled image (foreground) onto a labeled image (background), respectively. The two mixed images are fed into a Student network and supervised by the mixed supervisory signals of pseudo-labels and ground-truth. We reveal that the simple mechanism of copy-pasting bidirectionally between labeled and unlabeled data is good enough and the experiments show solid gains (e.g., over 21% Dice improvement on ACDC dataset with 5% labeled data) compared with other state-of-the-arts on various semi-supervised medical image segmentation datasets. Code is available at <https://github.com/DeepMed-Lab-ECNU/BCP>.

1. Introduction

Segmenting internal structures from medical images such as computed tomography (CT) or magnetic resonance imaging (MRI) is essential for many clinical applications [34]. Various techniques based on supervised learning for medical image segmentation have been proposed [4, 13, 45], which usually requires a large amount of labeled data. But, due to the tedious and expensive manual contouring process when labeling medical images, semi-supervised segmenta-

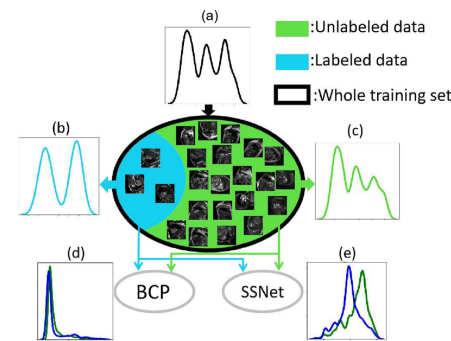


Figure 1. Illustration of the mismatch problem under semi-supervised learning setting. Assume the training set is drawn from a latent distribution in (a). But the empirical distributions of small amount of labeled data and a large amount of unlabeled data are (b) and (c), respectively. It’s hard to use few labeled data to construct the precise distribution of the whole dataset. (d) By using our BCP, the empirical distributions of labeled and unlabeled features are aligned. (e) But other methods such as SSNet [35] or cross unlabeled data copy-paste cannot address the empirical distribution mismatch issue. All distributions are kernel density estimations of voxels belonging to myocardium class in ACDC [2].

tion attracts more attention in recent years, and has become ubiquitous in the field of medical image analysis.

Generally speaking, in semi-supervised medical image segmentation, the labeled and unlabeled data are drawn from the same distribution, (Fig. 1 (a)). But in real-world scenario, it’s hard to estimate the precise distribution from labeled data because they are few in number. Thus, there always exists **empirical distribution mismatch** between a large amount of unlabeled and a very small amount of labeled data [30] (Fig. 1(b) and (c)). Semi-supervised segmentation methods always attempt to train labeled and unlabeled data symmetrically, in a consistent manner. E.g., self-training [1, 48] generates pseudo-labels to supervise unlabeled data in a pseudo-supervised manner. Mean Teacher based methods [40] adopt consistency loss to “supervise” unlabeled data with strong augmentations, in analogy with supervising labeled data with ground-truth. DTC [16] proposed a dual-task-consistency framework, applicable to

双向复制粘贴用于半监督医学图像分割

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摘要

在半监督医学图像分割中，有标注数据 and 无标注数据分布之间存在经验性的不匹配问题。如果将有标注数据和无标注数据分开处理或以不一致的方式处理，从有标注数据中获得的知識可能会被大量丢弃。我们提出了一种直接的方法来缓解该问题——双向复制粘贴有标注数据和无标注数据，采用简单的Mean Teacher架构。该方法鼓励无标注数据从有标注数据中学习全面的共同语义，涵盖向内和向外两个方向。更重要的是，有标注数据和无标注数据的一致学习过程可以大幅减小经验分布差距。具体而言，我们分别将有标注图像的随机裁剪区域（前景）复制粘贴到无标注图像（背景）上，以及将无标注图像（前景）复制粘贴到有标注图像（背景）上。这两张混合图像输入到学生网络中，并由伪标签和真实标签的混合监督信号进行监督。我们揭示了在有标注数据和无标注数据之间进行双向复制粘贴的简单机制已经足够好，实验表明，与其他最先进方法相比，在多个半监督医学图像分割数据集上取得了稳定的提升（例如，在仅使用5%有标注数据的ACDC数据集上，Dice提升超过21%）。代码可在 <https://github.com/DeepMed-Lab-ECNU/BCP> 获取。

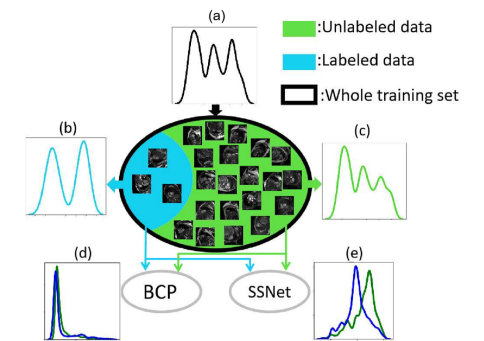


图1. 半监督学习设置下不匹配问题的图示。(a)中假设训练集从潜在分布中采样。但少量有标注数据和大量无标注数据的经验分布分别如(b)和(c)所示。很难用少量有标注数据来构建整个数据集的精确分布。(d)通过使用我们的BCP，有标注和无标注特征的经验分布得以对齐。(e)但其他方法如SSNet [35] 或跨无标注数据复制粘贴无法解决经验分布不匹配问题。所有分布均为ACDC [2]中心肌类别体素的核心密度估计。

近年来受到更多关注，并已在医学图像分析领域变得无处不在。

一般来说，在半监督医学图像分割中，有标注数据和无标注数据来自同一分布（图 1 (a)）。但在真实场景中，由于有标注数据数量很少，很难从中估计出精确的分布。因此，大量无标注数据与极少量有标注数据之间始终存在 [30] 经验分布不匹配1（图 (b) 和 (c)）。半监督分割方法总是试图以一致的方式对称地训练有标注数据和无标注数据。例如，自训练 [1, 48]生成伪标签，以伪监督方式监督无标注数据。基于 Mean Teacher 的方法 [40] 采用一致性损失，用强增强来“监督”无标注数据，类似于用真实标签监督有标注数据。DTC [16]提出了一个双任务一致性框架，适用于

1. 引言

从计算机断层扫描（CT）或磁共振成像（MRI）等医学图像中分割内部结构对于许多临床应用至关重要[34]。基于监督学习的各种医学图像分割技术已被提出 [4, 13, 45]，这些技术通常需要大量有标签数据。但由于标注医学图像时人工勾画过程繁琐且昂贵，半监督分割

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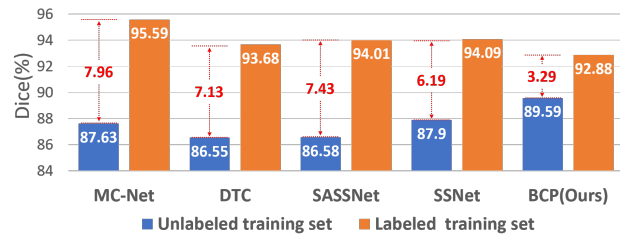


Figure 2. Dice scores for unlabeled and labeled training data of different models on LA dataset [39]. A much smaller performance gap is observed in our method.

both labeled and unlabeled data. ContrastMask [31] applied dense contrastive learning on both labeled and unlabeled data. But most existing semi-supervised methods used labeled and unlabeled data under separate learning paradigms. Thus, it often leads to the discarding of massive knowledge learned from the labeled data and the empirical distribution mismatch between labeled and unlabeled data (Fig. 1(e)).

CutMix [42] is simple yet strong data processing method, also dubbed as Copy-Paste (CP), which has the potential to encourage unlabeled data to learn common semantics from the labeled data, since pixels in the same map share semantics to be closer [29]. In semi-supervised learning, forcing consistency between weak-strong augmentation pair of unlabeled data is widely used [11, 14, 32, 47], and CP is usually used as a strong augmentation. But existing CP methods only consider CP cross unlabeled data [8, 10, 14], or simply copy crops from labeled data as foreground and paste to another data [6, 9]. They neglect to design a consistent learning strategy for labeled and unlabeled data, which hampers its usage on reducing the distribution gap. Meanwhile, CP tries to enhance the generalization of networks by increasing unlabeled data diversity, but a high performance is hard to achieve since CutMixed image is only supervised by low-precision pseudo-labels. It's intuitive to use more accurate supervision to help networks segment degraded region cut by CP.

To alleviate the empirical mismatch problem between labeled and unlabeled data, a successful design is to encourage unlabeled data to learn comprehensive common semantics from the labeled data, and meanwhile, furthering the distribution alignment via a consistent learning strategy for labeled and unlabeled data. We achieve this by proposing a **surprisingly simple yet very effective** Bidirectional Copy-Paste (BCP) method, instantiated in the Mean Teacher framework. Concretely, to train the Student network, we augment our inputs by copy-pasting random crops from a labeled image (foreground) onto an unlabeled image (background) and reversely, copy-pasting random crops from an unlabeled image (foreground) onto a labeled im-

age (background). The Student network is supervised by the generated supervisory signal via bidirectional copy-pasting between the pseudo-labels of the unlabeled images from the Teacher network and the label maps of the labeled image. The two mixed images help the network to learn common semantics between the labeled and unlabeled data **bidirectionally and symmetrically**. We compute the Dice scores for labeled and unlabeled training set from LA dataset [39] based on models trained by state-of-the-arts and our method, as shown in Fig. 2. Previous models which process labeled data and unlabeled data separately present strong performance gap between labeled and unlabeled data. *E.g.*, MC-Net obtains 95.59% Dice for labeled data but only 87.63% for unlabeled data. It means previous models absorb knowledge from ground-truth well, but discard a lot when transferring to unlabeled data. Our method can largely decrease the gap between labeled and unlabeled data (Fig. 1(d)) in terms of their performances. It is also interesting to observe that Dice for labeled data of our BCP is lower than other methods, implying that BCP can mitigate the over-fitting problem to some extent.

We verify BCP in three popular datasets: LA [39], Pancreas-NIH [21], and ACDC [2] datasets. Extensive experiments show our simple method outperforms all state-of-the-arts by a large margin, with even over 21% improvement in Dice on ACDC dataset with 5% labeled data. Ablation study further shows the effectiveness of each proposed module. Note that compared with the baseline *e.g.*, VNet or UNet, our method does not introduce new parameters for training, while remaining the same computational cost.

2. Related Work

2.1. Medical Image Segmentation

Segmenting internal structures from medical images is essential for many clinical applications [34]. Existing methods for medical image segmentation can be groups into two categories. The first category designed various 2D/3D segmentation network architectures [3, 4, 13, 18, 20, 49]. The second category leveraged medical prior knowledge to network training [23, 28, 33, 38].

2.2. Semi-supervised Medical Image Segmentation

Many efforts have been made in semi-supervised medical image segmentation. Entropy minimization (EM) and consistency regularization (CR) are the two widely-used loss functions. Meanwhile, many works extended Mean Teacher framework in different ways. SASSNet [12] leveraged unlabeled data to enforce a geometric shape constraint on the segmentation output. DTC [16] proposed a dual-task-consistency framework by building task-level regularization explicitly. SimCVD [40] modeled geometric structure and semantic information explicitly and constrain them

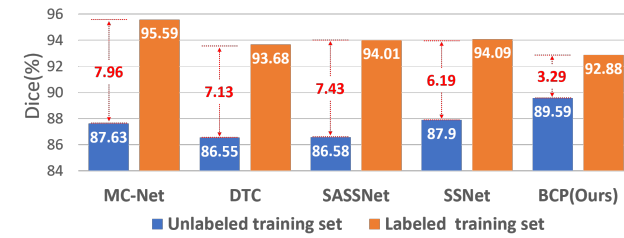


图2. 不同模型在LA数据集上对无标注和有标注训练数据的Dice分数 [39]。我们的方法观察到的性能差距要小得多。

有标注数据和无标注数据。ContrastMask [31] 对有标注数据和无标注数据应用了密集对比学习。但大多数现有半监督方法在有标注数据和无标注数据上采用分离的学习范式。因此，这往往导致从有标注数据中学到的大量知识被丢弃，以及有标注数据和无标注数据之间的经验分布不匹配（图 1（e））。

CutMix [42] 是一种简单但强大的数据处理方法，也被称为复制粘贴（CP），它有可能促使无标注数据从有标注数据中学习共同的语义，因为同一图像中的像素共享语义而更加接近 [29]。在半监督学习中，强制无标注数据的弱增强与强增强对之间的一致性被广泛使用 [11, 14, 32, 47]，而CP通常被用作一种强增强。但现有的CP方法仅考虑跨无标注数据的CP [8, 10, 14]，或者简单地从有标注数据中复制裁剪区域作为前景并粘贴到另一个数据上 [6, 9]。它们忽视了为有标注和无标注数据设计一致的学习策略，这阻碍了其在缩小分布差距方面的应用。同时，CP试图通过增加无标注数据的多样性来增强网络的泛化能力，但由于CutMix后的图像仅由低精度的伪标签监督，难以实现高性能。直观上，使用更准确的监督来帮助网络分割被CP裁剪的退化区域是合理的。

为了缓解有标签数据与无标签数据之间的经验不匹配问题，一种成功的设计是鼓励无标签数据从有标签数据中学习全面的共同语义，同时通过对有标签和无标签数据采用一致的学习策略来进一步促进分布对齐。我们通过提出一种**出奇简单却非常有效的双向复制粘贴（BCP）**方法来实现这一点，该方法在Mean Teacher框架中实例化。具体而言，为了训练学生网络，**我们将来自有标签图像的随机裁剪（前景）复制粘贴到无标签图像（背景）上来增强输入，反过来，将来自无标签图像的随机裁剪（前景）复制粘贴到有标签图**

像（背景）上。学生网络通过双向复制粘贴在来自教师网络的无标签图像的伪标签与有标签图像的标签图之间生成的监督信号来监督。这两张混合图像帮助网络双向且对称地学习有标签和无标签数据之间的共同语义。我们基于由最先进方法和我们的方法训练的模型，计算LA数据集中有标签和无标签训练集的Dice分数 [39]，如图2所示。先前分别处理有标签数据和无标签数据的模型在有标签和无标签数据之间表现出强烈的性能差距。例如，MC-Net在有标签数据上获得95.59%的Dice，但在无标签数据上仅为87.63%。这意味着先前模型能够很好地从真实标签中吸收知识，但在迁移到无标签数据时丢弃了大量知识。我们的方法可以在性能方面大幅缩小有标签和无标签数据之间的差距（图1（d））。同样有趣的是，我们的BCP在有标签数据上的Dice低于其他方法，这表明BCP能够在一定程度上缓解过拟合问题。

我们在三个常用数据集上验证BCP：LA [39]，Pancreas-NIH [21]，和ACDC [2]数据集。大量实验表明，我们的简单方法以较大优势超越了所有最先进方法，在仅使用5%有标签数据的ACDC数据集上，Dice指标提升甚至超过21%。消融研究进一步证明了每个提出模块的有效性。值得注意的是，与基线例如VNet或UNet相比，我们的方法未引入新的训练参数，同时保持了相同的计算成本。

2. 相关工作

2.1. 医学图像分割

从医学图像中分割内部结构对于许多临床应用至关重要 [34]。现有的医学图像分割方法可分为两类。第一类设计了各种2D/3D分割网络架构 [3, 4, 13, 18, 20, 49]。第二类利用医学先验知识进行网络训练 [23, 28, 33, 38]。

2.2. 半监督医学图像分割

在半监督医学图像分割方面已经做出了许多努力。**熵最小化（EM）和一致性正则化（CR）**是两种广泛使用的损失函数。同时，许多工作以不同方式扩展了Mean Teacher框架。SASSNet [12] 利用未标注数据对分割输出施加几何形状约束。DTC [16]通过显式构建任务级正则化提出了一种双任务一致性框架。SimCVD [40]显式建模几何结构和语义信息，并约束它们

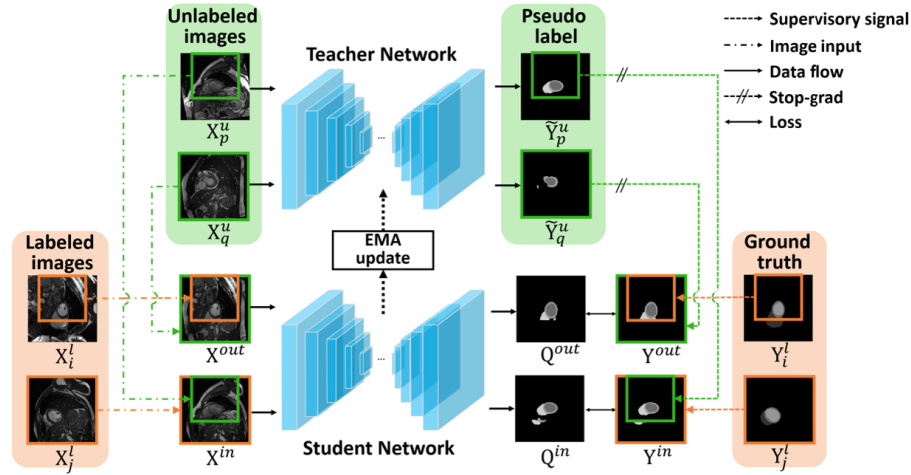


Figure 3. Overview of our bidirectional copy-paste framework in Mean Teacher architecture, drawn with 2D inputs for better visualization. The inputs to Student network are generated by mixing two labeled and two unlabeled images in the proposed bidirectional copy-paste manner. Then, to provide the supervisory signal to the Student network, we combine the ground-truths and the pseudo-labels generated by the Teacher network into one supervisory signal via the same bidirectional copy-paste, to enable strong supervision from ground-truths help the weak supervision from pseudo-label.

between Teacher and Student networks. These methods used geometric constraints to supervise the output of the network. UA-MT [41] used uncertainty information to guide Student network learn from the meaningful and reliable targets of Teacher network gradually. [46] combined image-wise and patch-wise representations to explore more complex similarity cues, enforcing the output to be consistent given different input sizes. CoraNet [22] proposed a model which can produce certain and uncertain regions, and Student network treats regions indicated from Teacher network with different weights. UMCT [37] used different perspectives of the network to predict the same image of different views. It utilized the predictions and the corresponding uncertainty to generate the pseudo-labels, which were used to supervise the prediction of unlabeled images. These methods have furthered the effectiveness for semi-supervised medical image segmentation. But, they ignored how to learn common semantics from labeled to unlabeled data. Treating labeled and unlabeled data separately often impedes knowledge transfer from labeled to unlabeled data.

2.3. Copy-Paste

Copy-paste is a simple but strong data processing method for many tasks, *e.g.*, instance segmentation [7, 9], semantic segmentation [6, 25] and object detection [5]. Generally speaking, copy-paste means copying crops of one image and pasting them onto another image. Mixup [43] and CutMix [42] are classic works of mixing whole images and mixing image crops respectively. Many recent works extended them to address specific goals. GuidedMix-Net [25] used mixup to generate higher-quality pseudo-labels

by transferring the knowledge of labeled data to the unlabeled data. InstaBoost [7] and Contextual Copy-Paste [5] placed the cropped foreground onto another image elaborately according to the surrounding visual context. CP² [27] proposed a pretraining method which copy-pastes a random crop from an image to another background image, it has been proved to be more suitable for downstream dense prediction tasks. [9] made a systematic study of copy-paste in instance segmentation. UCC [6] copied the pixels belonging to the class which has a low confidence score as foreground during training to alleviate the distribution mismatch and class imbalance problems. Previous methods only considered copy-paste cross unlabeled data, or simply copied crops from labeled data as foreground and pasted to another data. They neglect to design a consistent learning strategy for labeled and unlabeled data. Thus, a large distribution gap is still inevitable.

3. Method

Mathematically, we define the 3D volume of a medical image as $\mathbf{X} \in \mathbb{R}^{W \times H \times L}$. The goal of semi-supervised medical image segmentation is to predict the per-voxel label map $\hat{\mathbf{Y}} \in \{0, 1, \dots, K-1\}^{W \times H \times L}$, indicating where the background and the targets are in $\mathbf{X}$. K is the class number. Our training set $\mathcal{D}$ consists of N labeled data and M unlabeled data ($N \ll M$), expressed as two subsets: $\mathcal{D} = \mathcal{D}^l \cup \mathcal{D}^u$, where $\mathcal{D}^l = \{(\mathbf{X}_i^l, \mathbf{Y}_i^l)\}_{i=1}^N$ and $\mathcal{D}^u = \{\mathbf{X}_i^u\}_{i=N+1}^{M+N}$.

The overall pipeline of the proposed bidirectional copy-paste method is shown in Fig. 3, in the Mean Teacher architecture. We randomly pick two unlabeled images ($\mathbf{X}_p^u, \mathbf{X}_q^u$),

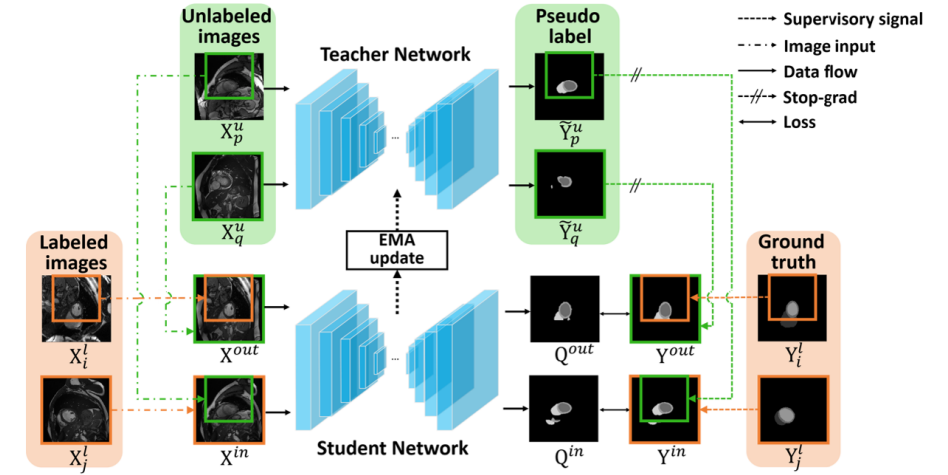


图3. 我们在Mean Teacher架构下的双向复制粘贴框架概览，为便于可视化采用2D输入绘制。学生网络的输入通过所提出的双向复制粘贴方式混合两张有标注图像和两张无标注图像生成。然后，为了向学生网络提供监督信号，我们通过相同的双向复制粘贴将真实标签和教师网络生成的伪标签合并为一个监督信号，从而使得来自真实标签的强监督能够帮助来自伪标签的弱监督。

在教师网络和学生网络之间。这些方法使用几何约束来监督网络的输出。UA-MT [41]使用不确定性信息引导学生网络逐步从教师网络中有意义且可靠的目标中学习。[46]结合了图像级和块级表示以探索更复杂的相似性线索，强制输出在不同输入尺寸下保持一致。CoraNet [22]提出了一种能够产生确定和不确定区域的模型，学生网络以不同权重对待教师网络指示的区域。UMCT [37]使用网络的不同视角来预测同一图像的不同视图。它利用预测结果及相应的不确定性来生成伪标签，用于监督无标注图像的预测。这些方法进一步提升了半监督医学图像分割的有效性。但是，它们忽略了如何从有标签数据到无标签数据学习共同语义。将有标签数据和无标签数据分开处理往往会阻碍从有标签数据到无标签数据的知识迁移。

2.3. 复制粘贴

复制粘贴是一种简单但强大的数据处理方法，适用于许多任务，例如实例分割 [7, 9]、语义分割 [6, 25] 和目标检测 [5]。一般来说，复制粘贴是指将一张图像的裁剪区域复制并粘贴到另一张图像上。Mixup [43]和CutMix [42]分别是混合整幅图像和混合图像裁剪区域的经典工作。近来许多工作对其进行了扩展以解决特定目标。GuidedMix-Net[25]使用mixup生成更高质量的伪标签

通过将有标注数据的知识迁移到无标注数据。

InstaBoost [7]和Contextual Copy-Paste [5]根据周围的视觉上下文精心地将裁剪的前景放置到另一张图像上。CP² [27]提出了一种预训练方法，将图像中的随机裁剪区域复制粘贴到另一个背景图像上，已被证明更适合下游密集预测任务。[9]对实例分割中的复制粘贴进行了系统研究。UCC [6]在训练过程中将属于置信度较低的类别的像素作为前景进行复制，以缓解分布不匹配和类别不平衡问题。先前的方法仅考虑了跨无标注数据的复制粘贴，或者简单地将有标注数据中的裁剪区域作为前景复制并粘贴到另一数据上。它们忽视了为有标注数据和无标注数据设计一致的学习策略。因此，较大的分布差距仍然不可避免。

3. 方法

在数学上，我们将医学图像的3D体积定义为 $\mathbf{X} \in \mathbb{R}^{W \times H \times L}$ 。半监督医学图像分割的目标是预测逐体素的标签图 $\hat{\mathbf{Y}} \in \{0, 1, \dots, K-1\}^{W \times H \times L}$ ，指示 $\mathbf{X}$ 中背景和目标的位置。 K 是类别数。我们的训练集 $\mathcal{D}$ 由 N 有标签数据和 M 无标签数据($N \ll M$)组成，表示为两个子集： $\mathcal{D} = \mathcal{D}^l \cup \mathcal{D}^u$ ，其中 $\mathcal{D}^l = \{(\mathbf{X}_i^l, \mathbf{Y}_i^l)\}_{i=1}^N$ 和 $\mathcal{D}^u = \{\mathbf{X}_i^u\}_{i=N+1}^{M+N}$ 。

所提出的双向复制粘贴方法的整体流程如图3所示，采用Mean Teacher架构。我们随机选取两张无标注图像 ($\mathbf{X}_p^u, \mathbf{X}_q^u$)，

and two labeled images ($\mathbf{X}_i^l, \mathbf{X}_j^l$) from the training set. Then we copy-paste a random crop from $\mathbf{X}_i^l$ (the foreground) onto $\mathbf{X}_q^u$ (the background) to generate the mixed image $\mathbf{X}^{out}$, and from $\mathbf{X}_p^u$ (the foreground) onto $\mathbf{X}_j^l$ (the background) to generate another mixed image $\mathbf{X}^{in}$. Unlabeled images are able to learn comprehensive common semantics from labeled images from both *inward* ($\mathbf{X}^{in}$) and *outward* ($\mathbf{X}^{out}$) directions. Images $\mathbf{X}^{in}$ and $\mathbf{X}^{out}$ are then fed into the Student network to predict segmentation masks $\hat{\mathbf{Y}}^{in}$ and $\hat{\mathbf{Y}}^{out}$. The segmentation masks are supervised by bidirectional copy-pasting the predictions of the unlabeled images from the Teacher network and the label maps of the labeled images.

3.1. Bidirectional Copy-Paste

3.1.1 Mean-Teacher and Training Strategy

In our BCP framework, there are a Teacher network, $\mathcal{F}_t(\mathbf{X}_p^u, \mathbf{X}_q^u; \Theta_t)$, and a Student network $\mathcal{F}_s(\mathbf{X}^{in}, \mathbf{X}^{out}; \Theta_s)$, where Θ_t and Θ_s are parameters. The Student network is optimized by stochastic gradient descent, and the Teacher network is by exponential moving average (EMA) of Student network. [24] Our training strategy is divided into three steps. We first use only labeled data to pretrain a model, then we use the pretrained model as Teacher network to generate pseudo-labels for unlabeled images. At each iteration, we first optimize the Student network parameters Θ_s by stochastic gradient descent. Finally, we update the Teacher network parameters Θ_t using EMA of the Student parameters Θ_s .

3.1.2 Pre-Training via Copy-Paste

Inspired by previous work [9], we conducted Copy-Paste augmentation on labeled data to train a supervised model, the supervised model will generate pseudo-labels for unlabeled data during self-training. This strategy was proved to be effective to improve segmentation performance, more details will be illustrated in ablation studies.

3.1.3 Bidirectional Copy-Paste Images

To conduct copy-paste between a pair of images, we first generate a zero-centered mask $\mathcal{M} \in \{0, 1\}^{W \times H \times L}$, indicating whether the voxel comes from the foreground (0) or the background (1) image. The size of the zero-value region is $\beta H \times \beta W \times \beta L$, where $\beta \in (0, 1)$. Then we bidirectionally copy-paste labeled and unlabeled images as follows:

$$\mathbf{X}^{in} = \mathbf{X}_j^l \odot \mathcal{M} + \mathbf{X}_p^u \odot (\mathbf{1} - \mathcal{M}), \quad (1)$$

$$\mathbf{X}^{out} = \mathbf{X}_q^u \odot \mathcal{M} + \mathbf{X}_i^l \odot (\mathbf{1} - \mathcal{M}), \quad (2)$$

where $\mathbf{X}_i^l, \mathbf{X}_j^l \in \mathcal{D}^l$, $i \neq j$, $\mathbf{X}_p^u, \mathbf{X}_q^u \in \mathcal{D}^u$, $p \neq q$, $\mathbf{1} \in \{1\}^{W \times H \times L}$, and $\odot$ means element-wise multiplication.

tion. Two labeled and unlabeled images are adopted to keep the diversity of the input.

3.1.4 Bidirectional Copy-Paste Supervisory Signals

To train the Student network, supervisory signals are also generated via BCP operation. Unlabeled images $\mathbf{X}_p^u$ and $\mathbf{X}_q^u$ are fed into the Teacher network, and their probability maps are computed:

$$\mathbf{P}_p^u = \mathcal{F}_t(\mathbf{X}_p^u; \Theta_t), \quad \mathbf{P}_q^u = \mathcal{F}_t(\mathbf{X}_q^u; \Theta_t). \quad (3)$$

The initial pseudo-label $\hat{\mathbf{Y}}^u$ (p and q are dropped for simplicity) are determined by taking a common threshold 0.5 on $\mathbf{P}^u$ for binary segmentation tasks, or taking *argmax* operation on $\mathbf{P}^u$ for multi-class segmentation tasks. The final pseudo-label $\tilde{\mathbf{Y}}^u$ is obtained via selecting the largest connected component of $\hat{\mathbf{Y}}^u$ which will effectively remove outlier voxels. Then, we propose to bidirectionally copy-paste the pseudo-labels of unlabeled images and ground truth labels of labeled images in the same manner as in Eq.1 and Eq.2 to obtain the supervisory signals:

$$\mathbf{Y}^{in} = \mathbf{Y}_j^l \odot \mathcal{M} + \tilde{\mathbf{Y}}_p^u \odot (\mathbf{1} - \mathcal{M}), \quad (4)$$

$$\mathbf{Y}^{out} = \tilde{\mathbf{Y}}_q^u \odot \mathcal{M} + \mathbf{Y}_i^l \odot (\mathbf{1} - \mathcal{M}). \quad (5)$$

$\mathbf{Y}^{in}$ and $\mathbf{Y}^{out}$ will be used as the supervision to supervise the Student network predictions of $\mathbf{X}^{in}$ and $\mathbf{X}^{out}$.

3.2. Loss Function

Each input image of the Student network consists of component from both labeled and unlabeled image. Intuitively, ground truth masks of labeled images are usually more accurate than pseudo-labels of unlabeled images. We use α to control the contribution of unlabeled image pixels to the loss function. The loss functions for $\mathbf{X}^{in}$ and $\mathbf{X}^{out}$ are computed respectively by

$$\mathcal{L}^{in} = \mathcal{L}_{seg}(\mathbf{Q}^{in}, \mathbf{Y}^{in}) \odot \mathcal{M} + \quad (6)$$

$$\begin{aligned} & \alpha \mathcal{L}_{seg}(\mathbf{Q}^{in}, \mathbf{Y}^{in}) \odot (\mathbf{1} - \mathcal{M}), \\ \mathcal{L}^{out} = & \mathcal{L}_{seg}(\mathbf{Q}^{out}, \mathbf{Y}^{out}) \odot (\mathbf{1} - \mathcal{M}) + \\ & \alpha \mathcal{L}_{seg}(\mathbf{Q}^{out}, \mathbf{Y}^{out}) \odot \mathcal{M}, \end{aligned} \quad (7)$$

where $\mathcal{L}_{seg}$ is the linear combination of Dice loss and Cross-entropy loss. $\mathbf{Q}^{in}$ and $\mathbf{Q}^{out}$ are computed by:

$$\mathbf{Q}^{in} = \mathcal{F}_s(\mathbf{X}^{in}; \Theta_s), \quad \mathbf{Q}^{out} = \mathcal{F}_s(\mathbf{X}^{out}; \Theta_s). \quad (8)$$

At each iteration we update the parameters Θ_s in Student network by stochastic gradient descent with the loss function:

$$\mathcal{L}_{all} = \mathcal{L}^{in} + \mathcal{L}^{out}. \quad (9)$$

以及训练集中的两张有标签图像 ($\mathbf{X}_i^l, \mathbf{X}_j^l$)。然后, 我们从 $\mathbf{X}_i^l$ (前景) 中随机裁剪一块区域并复制粘贴到 $\mathbf{X}_q^u$ (背景) 上, 生成混合图像 $\mathbf{X}^{out}$; 再从 $\mathbf{X}_p^u$ (前景) 复制粘贴到 $\mathbf{X}_j^l$ (背景) 上, 生成另一张混合图像 $\mathbf{X}^{in}$ 。无标注图像能够从有标签图像中学习到全面的共同语义, 涵盖向内 ($\mathbf{X}^{in}$) 和向外 ($\mathbf{X}^{out}$) 两个方向。随后, 图像 $\mathbf{X}^{in}$ 和 $\mathbf{X}^{out}$ 被输入学生网络, 以预测分割掩码 $\hat{\mathbf{Y}}^{in}$ 和 $\hat{\mathbf{Y}}^{out}$ 。分割掩码通过双向复制粘贴教师网络对无标注图像的预测结果以及有标签图像的标签图来进行监督。

3.1. 双向复制粘贴

3.1.1 均值教师与训练策略

在我们的BCP框架中, 有一个教师网络 $\mathcal{F}_t(\mathbf{X}_p^u, \mathbf{X}_q^u; \Theta_t)$, 以及一个学生网络 $\mathcal{F}_s(\mathbf{X}^{in}, \mathbf{X}^{out}; \Theta_s)$, 其中 Θ_t 和 Θ_s 是参数。学生网络通过随机梯度下降进行优化, 教师网络则通过学生网络的指数移动平均 (EMA) 进行优化。[24] 我们的训练策略分为三个步骤。我们首先仅使用有标签数据预训练一个模型, 然后我们将该预训练模型作为教师网络, 为无标签图像生成伪标签。在每次迭代中, 我们首先通过随机梯度下降优化学生网络参数 Θ_s 。最后, 我们使用学生参数的EMA更新教师网络参数 Θ_t Θ_s 。

3.1.2 通过复制粘贴进行预训练

受前人工作启发 [9], 我们对有标签数据进行复制粘贴增强以训练监督模型, 该监督模型将在自训练过程中为无标注数据生成伪标签。该策略已被证明能有效提升分割性能, 更多细节将在消融研究中说明。

3.1.3 双向复制粘贴图像

为在一对图像之间进行复制粘贴, 我们首先生成一个零中心掩码 $\mathcal{M} \in \{0, 1\}^{W \times H \times L}$, 指示体素来自前景 (0) 还是背景 (1) 图像。零值区域的大小为 $\beta H \times \beta W \times \beta L$, 其中 $\beta \in (0, 1)$ 。然后, 我们按如下方式对有标签图像和无标注图像进行双向复制粘贴:

$$\mathbf{X}^{in} = \mathbf{X}_j^l \odot \mathcal{M} + \mathbf{X}_p^u \odot (\mathbf{1} - \mathcal{M}), \quad (1)$$

$$\mathbf{X}^{out} = \mathbf{X}_q^u \odot \mathcal{M} + \mathbf{X}_i^l \odot (\mathbf{1} - \mathcal{M}), \quad (2)$$

其中 $\mathbf{X}_i^l, \mathbf{X}_j^l \in \mathcal{D}^l$, $i \neq j$, $\mathbf{X}_p^u, \mathbf{X}_q^u \in \mathcal{D}^u$, $p \neq q$, $\mathbf{1} \in \{1\}^{W \times H \times L}$ 和 $\odot$ 表示逐元素乘法

运算。采用两幅有标签图像和无标注图像以保持输入多样性。

3.1.4 双向复制粘贴监督信号

为训练学生网络, 监督信号也通过BCP操作生成。无标注图像 $\mathbf{X}_p^u$ 和 $\mathbf{X}_q^u$ 被输入教师网络, 并计算其概率图:

$$\mathbf{P}_p^u = \mathcal{F}_t(\mathbf{X}_p^u; \Theta_t), \quad \mathbf{P}_q^u = \mathcal{F}_t(\mathbf{X}_q^u; \Theta_t). \quad (3)$$

初始伪标签 $\hat{\mathbf{Y}}^u$ (p 以及 q 为简洁起见被省略) 的确定方式为: 在 $\mathbf{P}^u$ 上采用常用阈值0.5用于二分类分割任务, 或在 $\mathbf{P}^u$ 上执行 $\mathbf{argmax}$ 操作用于多类分割任务。最终的伪标签 $\tilde{\mathbf{Y}}^u$ 通过选取 $\hat{\mathbf{Y}}^u$ 的最大连通分量获得, 这将有效去除离群体素。然后, 我们提出以与公式1 和公式2 相同的方式, 双向复制粘贴无标注图像的伪标签和有标注图像的真值标签, 以获得监督信号:

$$\mathbf{Y}^{in} = \mathbf{Y}_j^l \odot \mathcal{M} + \tilde{\mathbf{Y}}_p^u \odot (\mathbf{1} - \mathcal{M}), \quad (4)$$

$$\mathbf{Y}^{out} = \tilde{\mathbf{Y}}_q^u \odot \mathcal{M} + \mathbf{Y}_i^l \odot (\mathbf{1} - \mathcal{M}). \quad (5)$$

$\mathbf{Y}^{in}$ 和 $\mathbf{Y}^{out}$ 将被用作监督信号, 以监督学生网络对 $\mathbf{X}^{in}$ 和 $\mathbf{X}^{out}$ 的预测。

3.2. 损失函数

学生网络的每张输入图像由来自有标注图像和无标注图像的组件构成。直观上, 有标注图像的真实掩码通常比无标注图像的伪标签更准确。我们使用 α 来控制无标注图像像素对损失函数的贡献。 $\mathbf{X}^{in}$ 和 $\mathbf{X}^{out}$ 的损失函数分别通过以下方式计算:

$$\mathcal{L}^{in} = \mathcal{L}_{seg}(\mathbf{Q}^{in}, \mathbf{Y}^{in}) \odot \mathcal{M} + \quad (6)$$

$$\begin{aligned} & \alpha \mathcal{L}_{seg}(\mathbf{Q}^{in}, \mathbf{Y}^{in}) \odot (\mathbf{1} - \mathcal{M}), \\ \mathcal{L}^{out} = & \mathcal{L}_{seg}(\mathbf{Q}^{out}, \mathbf{Y}^{out}) \odot (\mathbf{1} - \mathcal{M}) + \\ & \alpha \mathcal{L}_{seg}(\mathbf{Q}^{out}, \mathbf{Y}^{out}) \odot \mathcal{M}, \end{aligned} \quad (7)$$

其中 $\mathcal{L}_{seg}$ 是 Dice损失和交叉熵损失的线性组合。 $\mathbf{Q}^{in}$ 和 $\mathbf{Q}^{out}$ 通过以下方式计算得到:

$$\mathbf{Q}^{in} = \mathcal{F}_s(\mathbf{X}^{in}; \Theta_s), \quad \mathbf{Q}^{out} = \mathcal{F}_s(\mathbf{X}^{out}; \Theta_s). \quad (8)$$

在每次迭代中, 我们通过随机梯度下降使用损失函数更新学生网络内部的参数 Θ_s :

$$\mathcal{L}_{all} = \mathcal{L}^{in} + \mathcal{L}^{out}. \quad (9)$$

Method	Scans used		Metrics			
	Labeled	Unlabeled	Dice↑	Jaccard↑	95HD↓	ASD↓
V-Net	4(5%)	0	52.55	39.60	47.05	9.87
V-Net	8(10%)	0	82.74	71.72	13.35	3.26
V-Net	80(All)	0	91.47	84.36	5.48	1.51
UA-MT	4(5%)	76(95%)	82.26	70.98	13.71	3.82
SASSNet			81.60	69.63	16.16	3.58
DTC			81.25	69.33	14.90	3.99
URPC			82.48	71.35	14.65	3.65
MC-Net			83.59	72.36	14.07	2.70
SS-Net			86.33	76.15	9.97	2.31
Ours			88.02 ↑1.69	78.72 ↑2.57	7.90 ↓2.07	2.15 ↓0.16
UA-MT	8(10%)	72(90%)	87.79	78.39	8.68	2.12
SASSNet			87.54	78.05	9.84	2.59
DTC			87.51	78.17	8.23	2.36
URPC			86.92	77.03	11.13	2.28
MC-Net			87.62	78.25	10.03	1.82
SS-Net			88.55	79.62	7.49	1.90
Ours			89.62 ↑1.07	81.31 ↑1.69	6.81 ↓0.68	1.76 ↓0.14

Table 1. Comparisons with state-of-the-art semi-supervised segmentation methods on LA dataset. Improvements compared with the second best results are **highlighted**.

Afterwards, Teacher network parameters $\Theta_t^{(k+1)}$ at the $(k + 1)$ th iteration are updated:

$$\Theta_t^{(k+1)} = \lambda \Theta_t^{(k)} + (1 - \lambda) \Theta_s^{(k)}, \quad (10)$$

where λ is the smoothing coefficient parameter.

3.3. Testing Phase

In the testing stage, given a testing image $\mathbf{X}_{test}$, we obtain the probability map by: $\mathbf{Q}_{test} = \mathcal{F}(\mathbf{X}_{test}; \Theta_s)$, where $\hat{\Theta}_s$ are the well-trained Student network parameters. The final label map can be easily determined by $\mathbf{Q}_{test}$.

4. Experiments

4.1. Dataset

LA dataset. Atrial Segmentation Challenge [39] dataset includes 100 3D gadolinium-enhanced magnetic resonance image scans (GE-MRIs) with labels. We strictly follow the setting used in SSNet [35], DTC [16] and UA-MT [41]. **Pancreas-NIH.** Pancreas-NIH [21] dataset contains 82 contrast-enhanced abdominal CT volumes which are manually delineated. For fair comparison, we follow the setting in CoraNet [22].

ACDC dataset. ACDC [2] dataset is a four-class (*i.e.* background, right ventricle, left ventricle and myocardium) segmentation dataset, containing 100 patients’ scans. The data split [15] is fixed with 70, 10, and 20 patients’ scans for training, validation, and testing.

4.2. Evaluation Metrics

We choose four evaluation metrics: *Dice Score (%)*, *Jaccard Score (%)*, *95% Hausdorff Distance (95HD) in voxel*

Method	Scans used		Metrics			
	Labeled	Unlabeled	Dice↑	Jaccard↑	95HD↓	ASD↓
V-Net	12(20%)	50(80%)	69.96	55.55	14.27	1.64
DAN			76.74	63.29	11.13	2.97
ADVNET			75.31	61.73	11.72	3.88
UA-MT			77.26	63.82	11.90	3.06
SASSNet			77.66	64.08	10.93	3.05
DTC			78.27	64.75	8.36	2.25
CoraNet			79.67	66.69	7.59	1.89
Ours			82.91 ↑3.24	70.97 ↑4.28	6.43 ↓1.16	2.25↑0.61

Table 2. Comparisons with state-of-the-art semi-supervised segmentation methods on the Pancreas-NIH dataset.

Method	Scans used		Metrics			
	Labeled	Unlabeled	Dice↑	Jaccard↑	95HD↓	ASD↓
U-Net	3(5%)	0	47.83	37.01	31.16	12.62
U-Net	7(10%)	0	79.41	68.11	9.35	2.70
U-Net	70(All)	0	91.44	84.59	4.30	0.99
UA-MT	3(5%)	67(95%)	46.04	35.97	20.08	7.75
SASSNet			57.77	46.14	20.05	6.06
DTC			56.90	45.67	23.36	7.39
URPC			55.87	44.64	13.60	3.74
MC-Net			62.85	52.29	7.62	2.33
SS-Net			65.83	55.38	6.67	2.28
Ours			87.59 ↑21.76	78.67 ↑23.29	1.90 ↓4.77	0.67 ↓1.61
UA-MT	7(10%)	63(90%)	81.65	70.64	6.88	2.02
SASSNet			84.50	74.34	5.42	1.86
DTC			84.29	73.92	12.81	4.01
URPC			83.10	72.41	4.84	1.53
MC-Net			86.44	77.04	5.50	1.84
SS-Net			86.78	77.67	6.07	1.40
Ours			88.84 ↑2.06	80.62 ↑2.95	3.98 ↓2.09	1.17 ↓0.23

Table 3. Comparisons with state-of-the-art semi-supervised segmentation methods on the ACDC dataset.

and *Average Surface Distance (ASD) in voxel*. Given two object regions, Dice and Jaccard mainly compute the percentage of overlap between them, ASD computes the average distance between their boundaries, and 95HD measures the closest point distance between them.

4.3. Implementation Details

$\alpha = 0.5, \beta = 2/3$ are set as the default value in experiments, unless otherwise specified. We conduct all experiments on an NVIDIA 3090 GPU with fixed random seeds. **LA dataset.** Following SS-Net [35], we use rotation and flip operations to augment data and train our model via an SGD optimizer with the initial learning rate 0.01 decay by 10% every 2.5K iterations. The backbone is set as 3D V-Net. During training, we randomly crop $112 \times 112 \times 80$ patches, and the size of the zero-value region is $74 \times 74 \times 53$ ($\beta = 2/3$). The batch size is set as 8, containing four labeled patches and four unlabeled patches. The iterations of pre-training and self-training are set as 2k and 15k respectively.

Pancreas-NIH. Following CoraNet [22], we augment data by rotating, rescaling and flipping, and train a four-layer 3D

方法	使用的扫描		指标			
	有标注	无标注	Dice↑	Jaccard↑	95HD↓	ASD↓
V-Net	4(5%)	0	52.55	39.60	47.05	9.87
V-Net	8(10%)	0	82.74	71.72	13.35	3.26
V-Net	80(全部)	0	91.47	84.36	5.48	1.51
UA-MT	4(5%)	76(95%)	82.26	70.98	13.71	3.82
SASSNet			81.60	69.63	16.16	3.58
DTC			81.25	69.33	14.90	3.99
URPC			82.48	71.35	14.65	3.65
MC-Net			83.59	72.36	14.07	2.70
SS-Net			86.33	76.15	9.97	2.31
Ours			88.02 ↑1.69	78.72 ↑2.57	7.90 ↓2.07	2.15 ↓0.16
UA-MT	8(10%)	72(90%)	87.79	78.39	8.68	2.12
SASSNet			87.54	78.05	9.84	2.59
DTC			87.51	78.17	8.23	2.36
URPC			86.92	77.03	11.13	2.28
MC-Net			87.62	78.25	10.03	1.82
SS-Net			88.55	79.62	7.49	1.90
Ours			89.62 ↑1.07	81.31 ↑1.69	6.81 ↓0.68	1.76 ↓0.14

表1. 在LA数据集上与最先进半监督分割方法的比较。与次优结果相比的改进为 **highlighted**。

随后，教师网络参数 $\Theta_t^{(k+1)}$ 在第 $(k + 1)$ 次迭代时更新：

$$\Theta_t^{(k+1)} = \lambda \Theta_t^{(k)} + (1 - \lambda) \Theta_s^{(k)}, \quad (10)$$

其中 λ 是平滑系数参数。

3.3. 测试阶段

在测试阶段，给定一张测试图像 $\mathbf{X}_{test}$ ，我们通过以下方式获得概率图： $\mathbf{Q}_{test} = \mathcal{F}(\mathbf{X}_{test}; \Theta_s)$ ，其中 Θ_s 是训练良好的学生网络参数。最终的标签图可以通过 $\mathbf{Q}_{test}$ 轻松确定。

4. 实验

4.1. 数据集

LA数据集。心房分割挑战赛 [39] 数据集包含100例带有标签的三维钆增强磁共振图像扫描（GE-MRIs）。我们严格遵循SSNet [35], DTC [16]和UA-MT [41]中使用的设置。

Pancreas-NIH。Pancreas-NIH [21] 数据集包含 82例对比增强腹部CT体数据，均为人工勾画。为公平比较，我们遵循CoraNet [22]中的设置。

ACDC数据集。ACDC [2] 数据集是一个四类别（即背景、右心室、左心室和心肌）分割数据集，包含100例患者扫描。数据划分 [15] 固定为70、10和20例患者扫描，分别用于训练、验证和测试。

4.2. 评价指标

我们选择四个评估指标：*Dice*分数（%）、*Jaccard*分数（%）、体素单位的*95%豪斯多夫距离（95HD）*

方法	使用的扫描		指标			
	有标注	无标注	Dice↑	Jaccard↑	95HD↓	ASD↓
V-Net	12(20%)	50(80%)	69.96	55.55	14.27	1.64
DAN			76.74	63.29	11.13	2.97
ADVNET			75.31	61.73	11.72	3.88
UA-MT			77.26	63.82	11.90	3.06
SASSNet			77.66	64.08	10.93	3.05
DTC			78.27	64.75	8.36	2.25
CoraNet			79.67	66.69	7.59	1.89
Ours			82.91 ↑3.24	70.97 ↑4.28	6.43 ↓1.16	2.25↑0.61

表2. 在Pancreas-NIH数据集上与最先进半监督分割方法的比较。

方法	使用的扫描		指标			
	有标注	无标注	Dice↑	Jaccard↑	95HD↓	ASD↓
U-Net	3(5%)	0	47.83	37.01	31.16	12.62
U-Net	7(10%)	0	79.41	68.11	9.35	2.70
U-Net	70(全部)	0	91.44	84.59	4.30	0.99
UA-MT	3(5%)	67(95%)	46.04	35.97	20.08	7.75
SASSNet			57.77	46.14	20.05	6.06
DTC			56.90	45.67	23.36	7.39
URPC			55.87	44.64	13.60	3.74
MC-Net			62.85	52.29	7.62	2.33
SS-Net			65.83	55.38	6.67	2.28
Ours			87.59 ↑21.76	78.67 ↑23.29	1.90 ↓4.77	0.67 ↓1.61
UA-MT	7(10%)	63(90%)	81.65	70.64	6.88	2.02
SASSNet			84.50	74.34	5.42	1.86
DTC			84.29	73.92	12.81	4.01
URPC			83.10	72.41	4.84	1.53
MC-Net			86.44	77.04	5.50	1.84
SS-Net			86.78	77.67	6.07	1.40
Ours			88.84 ↑2.06	80.62 ↑2.95	3.98 ↓2.09	1.17 ↓0.23

表 3. 在 ACDC 数据集上与最先进的半监督分割方法的比较。

以及体素单位的平均表面距离（*ASD*）。给定两个目标区域，Dice和Jaccard主要计算它们之间的重叠百分比，ASD计算它们边界之间的平均距离，而95HD衡量它们之间的最近点距离。

4.3. 实现细节

$\alpha = 0.5, \beta = 2/3$ 除非另有说明，否则在实验中设为默认值。我们在配备固定随机种子的NVIDIA 3090 GPU上进行所有实验。 **LA数据集。**遵循 SS-Net [35],，我们使用旋转和翻转操作来增强数据，并通过SGD优化器训练模型，初始学习率为0.01，每 2.5K次迭代衰减10%。骨干网络设为3D V-Net。训练期间，我们随机裁剪 $112 \times 112 \times 80$ 图像块，零值区域的大小为 $74 \times 74 \times 53$ ($\beta = 2/3$)。批量大小设为 8，包含四个有标签图像块和四个无标签图像块。预训练和自训练的迭代次数分别设为2k和15k。

Pancreas-NIH。遵循CoraNet [22],，我们通过旋转、缩放和翻转来增强数据，并训练一个四层3D

Method	LA						ACDC					
	Scans used		Metrics				Scans used		Metrics			
	Labeled	Unlabeled	Dice↑	Jaccard↑	95HD↓	ASD↓	Labeled	Unlabeled	Dice↑	Jaccard↑	95HD↓	ASD↓
In	4(5%)	76(95%)	87.35	77.77	8.75	2.21	3(5%)	67(95%)	81.68	70.07	4.69	1.28
Out			87.32	77.78	9.38	2.16			72.19	60.69	39.57	18.15
CP			79.67	67.05	14.66	3.21			81.80	71.70	16.29	6.43
Ours			88.02	78.72	7.90	2.15			87.59	78.67	1.90	0.67
In	8(10%)	72(90%)	89.02	80.38	8.08	1.81	7(10%)	63(90%)	85.55	75.65	4.93	1.50
Out			87.61	78.10	8.99	2.63			87.23	78.07	8.61	2.39
CP			86.74	77.18	8.65	2.26			88.17	79.64	6.14	1.45
Ours			89.62	81.31	6.81	1.76			88.84	80.62	3.98	1.17

Table 4. Ablation study of the copy-paste directions. In: *inward* copy-paste (foreground: unlabeled, background: labeled). Out: *outward* copy-paste (foreground: labeled, background: unlabeled). CP: *direct* copy-paste (background & foreground: labeled & labeled and unlabeled & unlabeled).

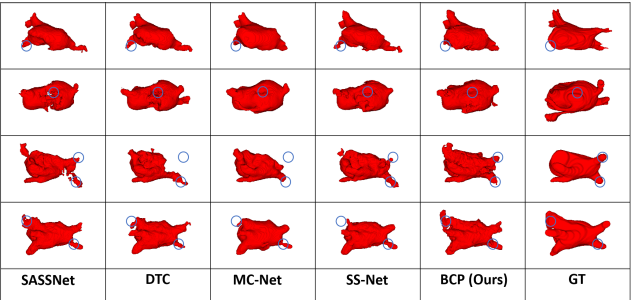


Figure 4. Visualizations of several semi-supervised segmentation methods with 10% labeled data and ground truth on LA dataset (best viewed by zoom-in on screen).

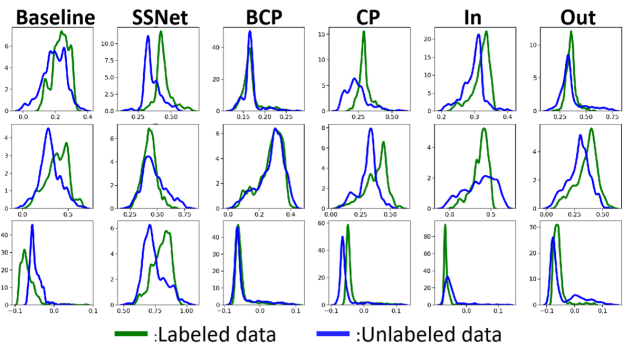


Figure 5. Kernel dense estimations of different methods, trained on 10% labeled ACDC dataset. Top to bottom are kernel dense estimations of features belong to three different class of ACDC: right ventricle, myocardium and left ventricle. Baseline: Only labeled data are used to train the network. *CP*, *In* and *Out* are same as Table 4. It can be seen that our BCP could make the features of labeled data and unlabeled data align better. Furthermore, the outstanding performance of our method compared with *In* and *Out* demonstrates the necessity of *bidirectional* copy-paste.

V-Net by Adam optimizer with initial learning rate as 0.001. During training, we randomly crop $96 \times 96 \times 96$ patches input the network, the size of the zero-value region of mask $\mathcal{M}$ is $64 \times 64 \times 64$. We set the batch size, pre-training epochs and the self-training epochs as 8, 60 and 200 respectively. **ACDC dataset.** Following SS-Net [35], we use 2D U-Net

Method	Scans used		Metrics			
	Labeled	Unlabeled	Dice↑	Jaccard↑	95HD↓	ASD↓
Mixup	4(5%)	76(95%)	41.71	29.58	59.75	21.87
FG-CutMix			67.05	54.00	30.52	6.15
Ours			88.02	78.72	7.90	2.15
Mixup	8(10%)	72(90%)	63.64	52.51	21.67	3.61
FG-CutMix			83.58	72.70	11.96	2.56
Ours			89.62	81.31	6.81	1.76

Table 5. Ablation study of interpolation strategies on LA dataset. Mixup: We imitate the framework of GuidedMix-Net [25], which is proposed for semi-supervised segmentation of natural images. FG-CutMix: We crop images of the whole training batch into 4×4 patches and then combine them randomly to generate new images.

as the backbone of our experiments. During training, the input patch size is 256×256 (2D slices) and the size of the zero-value region of mask $\mathcal{M}$ is 170×170 . The batch size, pre-training iterations, and the self-training training iterations are set as 24, 10k and 30k respectively.

4.4. Comparison with Sate-of-the-Art Methods

LA dataset We compare our framework on LA dataset with various competitors: UA-MT [41], SASSNet [12], DTC [16], URPC [17], MC-Net [36] and SS-Net [35]. Following SS-Net, semi-supervised experiments of different labeled ratios (*i.e.* 5% and 10%) are carried out. Results from other competitors were reported in the identical experimental setting in SS-Net [35] for fair comparisons. As shown in Table 1, our method achieves the best performance on all four evaluation metrics, outperforming other competitors by a big margin. Thanks to BCP, the network “sees” more variances for boundary regions or semantic change of voxels, allowing for achieving good shape-related performances (see 95HD and ASD) without any explicit boundary or shape constraints during training. Moreover, it can be seen in Fig. 4 that our method can segment fine details of the target organ, especially edges that are easy to be misidentified (the first row) or missed (the second, third and fourth row), highlighted by blue circles.

Pancreas-NIH dataset We conduct experiments on Pancreas-NIH dataset with 20% labeled ratio [16, 22]. We

方法	LA						ACDC					
	使用的扫描		指标				使用的扫描		指标			
	有标注	无标注	Dice↑	Jaccard↑	95HD↓	ASD↓	有标注	无标注	Dice↑	Jaccard↑	95HD↓	ASD↓
In	4(5%)	76(95%)	87.35	77.77	8.75	2.21	3(5%)	67(95%)	81.68	70.07	4.69	1.28
Out			87.32	77.78	9.38	2.16			72.19	60.69	39.57	18.15
CP			79.67	67.05	14.66	3.21			81.80	71.70	16.29	6.43
Ours			88.02	78.72	7.90	2.15			87.59	78.67	1.90	0.67
In	8(10%)	72(90%)	89.02	80.38	8.08	1.81	7(10%)	63(90%)	85.55	75.65	4.93	1.50
Out			87.61	78.10	8.99	2.63			87.23	78.07	8.61	2.39
CP			86.74	77.18	8.65	2.26			88.17	79.64	6.14	1.45
Ours			89.62	81.31	6.81	1.76			88.84	80.62	3.98	1.17

表 4. 复制粘贴方向的消融研究。In： 向内复制粘贴（前景：无标注，背景：有标注）。Out： 向外复制粘贴（前景：有标注，背景：无标注）。CP： 直接复制粘贴（背景和前景：有标注和有标注以及无标注和无标注）。

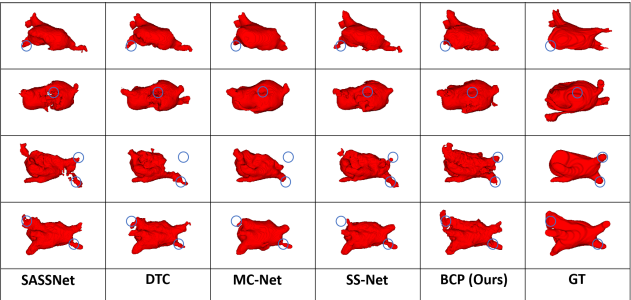


图4. 在LA数据集上使用10%有标签数据时几种半监督分割方法与真值的可视化对比（建议在屏幕上放大查看）。

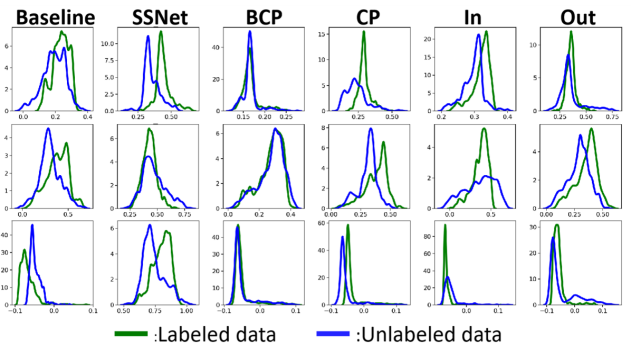


图5. 不同方法的核密度估计，在10%有标签ACDC数据集上训练。从上到下分别为ACDC三个不同类别特征的核密度估计：右心室、心肌和左心室。基线：仅使用有标注数据训练网络。*CP*、内部和外部与表4相同。可以看出，我们的BCP能够使有标签数据 and 无标签数据的特征更好地对齐。此外，与内部和外部相比，我们方法的优异性能证明了双向复制粘贴的必要性。

V-Net，使用Adam优化器，初始学习率设为0.001。训练期间，我们随机裁剪 $96 \times 96 \times 96$ 图像块输入网络，掩码的零值区域大小 $\mathcal{M}$ 为 $64 \times 64 \times 64$ 。我们将批量大小、预训练轮次和自训练轮次分别设为8、60和200。

ACDC数据集. 遵循SS-Net [35], 我们使用2D U-Net

方法	使用的扫描		指标			
	有标注	无标注	Dice↑	Jaccard↑	95HD↓	ASD↓
Mixup	4 (5%)	76(95%)	41.71	29.58	59.75	21.87
FG-CutMix			67.05	54.00	30.52	6.15
Ours			88.02	78.72	7.90	2.15
Mixup	8(10%)	72(90%)	63.64	52.51	21.67	3.61
FG-CutMix			83.58	72.70	11.96	2.56
Ours			89.62	81.31	6.81	1.76

表 5. 在LA数据集上插值策略的消融研究。Mixup： 我们模仿了GuidedMix-Net [25], 的框架， 该框架是为自然图像的半监督分割而提出的。FG-CutMix： 我们将整个训练批次的图像裁剪成 4×4 图像块， 然后随机组合它们以生成新图像。

作为我们实验的骨干网络。训练过程中，输入图像块尺寸为 256×256 （二维切片），掩码 $\mathcal{M}$ 的零值区域尺寸为 170×170 。批量大小、预训练迭代次数和自训练迭代次数分别设置为24、10k和30k。

4.4. 与最先进方法的比较

LA数据集 我们在LA数据集上将我们的框架与各种竞争方法进行比较: UA-MT [41], SASSNet [12],DTC [16], URPC [17], MC-Net [36]和SS-Net [35]。遵循SS-Net，我们开展了不同标注比例（即 5%和10%）的半监督实验。其他竞争方法的结果在SS-Net中相同的实验设置下报告 [35]，以进行公平比较。如表1所示，我们的方法在所有四个评价指标上均取得了最佳性能，大幅优于其他竞争方法。得益于BCP，网络能够“看到”边界区域或体素语义变化的更多差异，从而在训练期间无需任何显式的边界或形状约束即可实现良好的形状相关性能（参见95HD和ASD）。此外，从图4 中可以看出，我们的方法能够分割目标器官的精细细节，尤其是那些容易被误识别（第一行）或遗漏（第二、三、四行）的边缘，这些边缘由蓝色圆圈突出显示。

Pancreas-NIH数据集我们在Pancreas-NIH数据集上以20%的标注比例进行实验 [16, 22]。我们

Mode	Scans used		Metrics			
	Labeled	Unlabeled	Dice↑	Jaccard↑	95HD↓	ASD↓
Random	4(5%)	76(95%)	86.15	76.03	9.19	2.38
Contact			86.64	76.32	9.61	2.58
Ours			88.02	78.72	7.90	2.15
Random	8(10%)	72(90%)	84.50	73.79	10.79	2.39
Contact			88.66	79.82	7.93	2.27
Ours			89.62	81.31	6.81	1.76

Table 6. Results with three masking strategies on LA dataset.

β	Scans used		Metrics			
	Labeled	Unlabeled	Dice↑	Jaccard↑	95HD↓	ASD↓
1/3	4(5%)	76(95%)	79.92	67.73	15.44	3.63
1/2			86.49	76.63	8.74	2.23
2/3			88.02	78.72	7.90	2.15
5/6			87.92	78.57	8.29	2.26
1/3			83.20	72.04	11.64	2.94
1/2	8(10%)	72(90%)	88.81	89.10	7.33	1.96
2/3			89.62	81.31	6.81	1.76
5/6			88.75	79.96	7.63	2.07

Table 7. Ablation study of β on LA dataset.

compared BCP with V-Net [19], DAN [44], ADVNET [26], UA-MT [41], SASSNet [12], DTC [16] and CoraNet [22] in Table 2. In this table, DAN, ADVNET, UA-MT, SASSNet, DTC, CoraNet and our method took both labeled and unlabeled data to train the network with V-Net as the backbone, while V-Net only uses labeled data in the supervised setting (lower bound). BCP achieves significant improvement on Dice, Jaccard and 95HD (*i.e.*, surpassing the second best by 3.24%, 4.28% and 1.16, respectively). These results do not conduct any post-processing for fair comparison.

ACDC dataset Table 3 shows the averaged performance of four-class segmentation results on ACDC dataset with 5% and 10% labeled ratios. BCP surpasses all state-of-the-arts. We obtain a huge performance improvement up to 21.76% in terms of Dice for the setting with 5% labeled ratio. Following SS-Net [35], 2D slices are used to train our network. Noted that one 3D volume can be sliced into many 2D slices, so much more combinations from labeled and unlabeled slices could be produced than those using 3D data. Hence, during training, the knowledge of labeled data can be transferred to unlabeled data more sufficiently, especially when the number of labeled volume is very small. This might be the reason for such a significant improvement compared with others when the labeled ratio is 5%.

4.5. Ablation Studies

We conduct ablation studies to show the impact of each component in BCP. Including CP directions, design choices of masking strategies, interpolation strategies, β (size ratio for zero-value region in $\mathcal{M}$), and α (Eq. 6-7). We also investigate step-by-step the significant improvement of our method compared with competitors on the ACDC dataset with 5% labeled ratio. Some ablation studies on ACDC dataset are shown in the supplementary material.

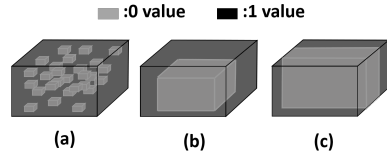


Figure 6. Different masking strategies. (a): random mask; (b): zero-centered mask; (c): contact mask.

Copy-Paste Direction We design three experiments to investigate the influence of different copy-paste directions in Table 4. *Inward* and *outward* copy-paste (*In* and *Out* in the table) mean using $\mathbf{X}^l \odot \mathcal{M} + \mathbf{X}^u \odot (\mathbf{1} - \mathcal{M})$ or $\mathbf{X}^u \odot \mathcal{M} + \mathbf{X}^l \odot (\mathbf{1} - \mathcal{M})$ respectively to train the network. We also conduct within-set copy-paste (*CP* in the table), *i.e.*, copy-paste labeled data on another labeled data and copy-paste unlabeled data on another unlabeled data. We can see that all these variants get inferior performances compared with our BCP, since they either lack consistent manner for training labeled and unlabeled data, or lack common semantics transfer between labeled and unlabeled data.

Interpolation Strategies We compare BCP with other two interpolation strategies: Mixup [43] and Fine-Grained CutMix (FG-CutMix). For Mixup, we superimpose labeled and unlabeled data to generate new training images, imitating the framework of GuidedMix-Net [25]. For FG-CutMix, we crop training images into 4×4 patches and combine them in batch to generate new images. The network predictions of new images are re-combined and then supervised by ground-truth or pseudo-labels. The results of LA dataset are shown in Table 5. Due to similar spatial structures of medical images, Mixup brings more influential noise in medical images. FG-CutMix maintains less structure information after CutMix than BCP. More details will be discussed in supplementary material.

Design Choices of Masking Strategies As shown in Fig. 6, we explore different masking strategies in BCP on LA dataset. To conduct a fair comparison, we maintain the same number of zero-value voxels for different strategies. We randomly sample 27 small $\beta H \times \beta W \times \beta L$ zero-value cubes in an all-one mask, and set $\beta = 2/9$ for each zero-value cube. For contact mask, the shape of zero-value region is $\beta H \times W \times L$, and β is set as 8/27 to control the number of zero-value voxels. As shown in Table 6, random mask obtains worst performance, since small random cubes only contain incoherently local foreground of an image, which lacks the ability in learning complete foreground representation. Contact mask has better integrity of foreground information, which performs better than random mask, but still performs worse than zero-centered mask used in our method, since foreground has less chance interacting with the background, compared with zero-centered mask. Thus,

Mode	使用的扫描		指标			
	有标注	无标注	Dice↑	Jaccard↑	95HD↓	ASD↓
随机	4(5%)	76(95%)	86.15	76.03	9.19	2.38
接触			86.64	76.32	9.61	2.58
Ours			88.02	78.72	7.90	2.15
随机	8(10%)	72(90%)	84.50	73.79	10.79	2.39
接触			88.66	79.82	7.93	2.27
Ours			89.62	81.31	6.81	1.76

表6. 在LA数据集上使用三种掩码策略的结果。

β	使用的扫描		指标			
	有标注	无标注	Dice↑	Jaccard↑	95HD↓	ASD↓
1/3	4(5%)	76(95%)	79.92	67.73	15.44	3.63
1/2			86.49	76.63	8.74	2.23
2/3			88.02	78.72	7.90	2.15
5/6			87.92	78.57	8.29	2.26
1/3			83.20	72.04	11.64	2.94
1/2	8(10%)	72(90%)	88.81	89.10	7.33	1.96
2/3			89.62	81.31	6.81	1.76
5/6			88.75	79.96	7.63	2.07

表 7. 在LA数据集上 β 的消融研究。

将BCP与V-Net [19], DAN [44], ADVNET [26], UA-MT [41], SASSNet [12], DTC [16] 以及CoraNet [22] 在表2中进行了比较。在该表中, DAN、ADVNET、UA-MT、SASSNet、DTC、CoraNet以及我们的方法均采用有标注数据和无标注数据以V-Net作为骨干网络来训练网络, 而V-Net在有监督设置下仅使用有标注数据(下界)。BCP在Dice、Jaccard和95HD上取得了显著提升(即, 分别比第二优的方法高出3.24%、4.28%和1.16)。为公平比较, 这些结果未进行任何后处理。

ACDC数据集表3展示了在ACDC数据集上标注比例为5%和10%时的四类分割平均性能。BCP超越了所有最先进的方法。在标注比例为5%的设置下, 我们在Dice指标上获得了高达21.76%的巨大性能提升。遵循SS-Net [35], 2, 我们使用D切片来训练网络。需要注意的是, 一个3D体积可以被切分成许多二维切片, 因此相比于使用3D数据, 可以从有标注和无标注切片中产生更多的组合。因此, 在训练过程中, 有标注数据的知识可以更充分地迁移到无标注数据, 尤其是在有标注体积数量非常少的情况下。这可能是在标注比例为5%时相比其他方法取得如此显著提升的原因。

4.5. 消融研究

我们进行消融研究以展示BCP中每个组件的影响。包括复制粘贴方向、掩码策略的设计选择、插值策略、 β (零值区域的大小比例在 $\mathcal{M}$ 中), 以及 α (公式 6-7)。我们还逐步研究了我们的方法在5%标注比例的ACDC数据集上相较于竞争方法的显著提升。一些在ACDC数据集上的消融研究展示在补充材料中。

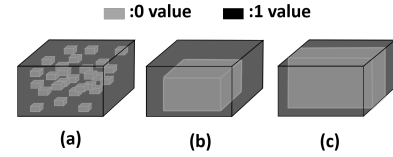


图 6. 不同的掩码策略。(a): 随机掩码; (b): 零中心掩码; (c): 接触掩码。

复制粘贴方向 我们设计了三个实验来研究表4中不同复制粘贴方向的影响。向内和向外复制粘贴(表中*In*和*Out*)分别表示使用 $\mathbf{X}^l \odot \mathcal{M} + \mathbf{X}^u \odot (\mathbf{1} - \mathcal{M})$ 或 $\mathbf{X}^u \odot \mathcal{M} + \mathbf{X}^l \odot (\mathbf{1} - \mathcal{M})$ 来训练网络。我们还进行了集合内复制粘贴(表中*CP*), 即, 将有标注数据复制粘贴到另一个有标注数据上, 以及将无标注数据复制粘贴到另一个无标注数据上。我们可以看到, 所有这些变体相较于我们的BCP都获得了较差的性能, 因为它们要么缺乏对有标注和无标注数据训练的一致方式, 要么缺乏有标注和无标注数据之间的共同语义迁移。

插值策略 我们将BCP与其他两种插值策略进行比较: Mixup [43]和细粒度CutMix (FG-CutMix)。对于Mixup, 我们叠加有标注和无标注数据以生成新的训练图像, 模仿GuidedMix-Net的框架 [25]。对于FG-CutMix, 我们将训练图像裁剪为 4×4 图像块并按批次组合以生成新图像。新图像的网络预测被重新组合, 然后由真实标签或伪标签进行监督。LA数据集的结果如表5所示。由于医学图像具有相似的空间结构, Mixup在医学图像中带来了更具影响力的噪声。FG-CutMix在CutMix后保留的结构信息比BCP更少。更多细节将在补充材料中讨论。

掩码策略的设计选择 如图6所示, 我们在LA数据集上探索了BCP中不同的掩码策略。为了进行公平比较, 我们对不同策略保持相同数量的零值体素。我们在全一掩码中随机采样27个小的 $\beta H \times \beta W \times \beta L$ 零值立方体, 并为每个零值立方体设置 $\beta = 2/9$ 。对于接触掩码, 零值区域的形状为 $\beta H \times W \times L$, 并将 β 设为 8/27以控制零值体素的数量。如表6所示, 随机掩码取得了最差的表现, 因为小的随机立方体仅包含图像中不连贯的局部前景, 缺乏学习完整前景表征的能力。接触掩码具有更好的前景信息完整性, 其表现优于随机掩码, 但仍不如我们方法中使用的零中心掩码, 因为与零中心掩码相比, 前景与背景交互的机会更少。因此,

α	LA						ACDC					
	Scans used		Metrics				Scans used		Metrics			
	Labeled	Unlabeled	Dice↑	Jaccard↑	95HD↓	ASD↓	Labeled	Unlabeled	Dice↑	Jaccard↑	95HD↓	ASD↓
0.5	4(5%)	76(95%)	88.02	78.72	7.90	2.15	3(5%)	67(95%)	87.59	78.67	1.90	0.67
1.5			87.21	77.49	8.67	2.37			85.88	76.02	3.17	0.93
2.5			86.56	76.46	9.82	2.60			85.43	75.47	12.02	4.05
0.5	8(10%)	72(90%)	89.62	81.31	6.81	1.76	7(10%)	63(90%)	88.84	80.62	3.98	1.17
1.5			89.35	80.88	7.46	2.09			88.65	80.31	1.99	0.68
2.5			88.74	79.88	7.73	2.15			87.13	78.19	3.67	1.24

Table 8. Ablation study of the weights α in the loss function.

Strategy	Scans used		Metrics			
	Labeled	Unlabeled	Dice↑	Jaccard↑	95HD↓	ASD↓
random	4(5%)	76(95%)	86.06	75.96	9.48	2.33
w/o CP			86.46	76.50	8.93	2.31
Ours			88.02	78.72	7.90	2.15
random	8(10%)	72(90%)	87.93	78.67	8.24	2.08
w/o CP			88.75	79.88	7.66	1.83
Ours			89.62	81.31	6.81	1.76

Table 9. Ablation study of pre-training strategy on LA dataset. random: Initialized randomly. w/o CP: Initialized from a pre-trained model trained on labeled data without copy-paste.

BCP	nms	Pre-Train	Dice↑	Jaccard↑	95HD↓	ASD↓
			47.62	36.61	29.02	11.46
✓			83.26	72.71	23.90	7.49
✓	✓		82.33	72.76	9.78	4.74
✓	✓	✓	87.59	78.67	1.90	0.67

Table 10. Ablation on ACDC dataset with 5% labeled data, $\alpha = 0.5$ across all experiments. nms: Post-processing the pseudo-labels for unlabeled data. Pre-Train: Initialized from a pre-trained model with copy-paste on labeled data.

both random mask and contact mask have weaker ability in mitigating the distribution mismatch problem between labeled and unlabeled data.

Size of Zero-value Region in M We study the impact of zero-value region size on LA dataset, as shown in the Table 7. For the zero-value region $\beta H \times \beta W \times \beta L$ in the mask $\mathcal{M}$, we set $\beta = \{\frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{5}{6}\}$. The performance gets worse as β decreases, which means small copy-pasted foreground has limited ability in transferring common semantics to/from the background. Best performance is achieved when $\beta = 2/3$ and it decreases a bit when $\beta = 5/6$.

Weight in Loss Function We set $\alpha = 0.5$ as the default value. Now we vary $\alpha = \{0.5, 1.5, 2.5\}$ to see how the performance changes. Table 8 shows it is not sensitive when α changes from 0.5 to 1.5, but an obvious performance drop is observed when $\alpha = 2.5$.

Teacher Network Initialization Strategy In our default setting, the Teacher network is initialized by a pre-trained model, which is trained on labeled data in a copy-paste manner. We study other network initialization strategies: initialized randomly and initialized from a pre-trained model which is trained on labeled data without copy-paste. Comparison results are shown in Table 9. Compared with pre-

training on original labeled data, performing copy-paste on the labeled data during pre-train can effectively improve the generalization ability of the network.

Ablation on ACDC with 5% Labeled Data In Table 3, our method achieves a huge improvement on ACDC dataset with 5% labeled data. We separate our method into three components and study which component contributes the most to this improvement. As shown in Table 10, without our component (the first row), it degenerated into a normal pseudo-label-based self-training method, which means the segmentation of labeled and unlabeled images are supervised by ground truth and pseudo-labels respectively. Then, BCP leads to a significant performance gain (from 47.62% to 83.26% in Dice). Post-processing (nms) and a better Teacher network initialization enhance the quality of pseudo-labels and thus further improve the performance.

5. Conclusion

We have presented the bidirectional copy-paste (BCP) for semi-supervised medical image segmentation. We extend copy-paste-based method in a bidirectional manner, which reduces the distribution gap between labeled and unlabeled data. Experiments on LA, NIH-Pancreas and ACDC datasets show the superiority of the proposed BCP, with even over 21% Dice improvement on ACDC dataset with 5% labeled data. Note that BCP does not introduce new parameters or computational cost compared with the backbone network. **Limitations.** We didn’t specifically design a module to enhance local attributes learning. Though BCP performs better than all competitors, target parts with extremely low contrast are still hard to segment well (*e.g.*, bottom left part on 2nd row of Fig. 4 is missing).

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α	LA						ACDC					
	使用的扫描		指标				使用的扫描		指标			
	有标注	无标注	Dice↑	Jaccard↑	95HD↓	ASD↓	有标注	无标注	Dice↑	Jaccard↑	95HD↓	ASD↓
0.5	4(5%)	76(95%)	88.02	78.72	7.90	2.15	3(5%)	67(95%)	87.59	78.67	1.90	0.67
1.5			87.21	77.49	8.67	2.37			85.88	76.02	3.17	0.93
2.5			86.56	76.46	9.82	2.60			85.43	75.47	12.02	4.05
0.5	8(10%)	72(90%)	89.62	81.31	6.81	1.76	7(10%)	63(90%)	88.84	80.62	3.98	1.17
1.5			89.35	80.88	7.46	2.09			88.65	80.31	1.99	0.68
2.5			88.74	79.88	7.73	2.15			87.13	78.19	3.67	1.24

表 8. 损失函数中权重 α 的消融研究。

策略	使用的扫描		指标			
	有标注	无标注	Dice↑	Jaccard↑	95HD↓	ASD↓
随机 无复制粘贴	4(5%)	76(95%)	86.06	75.96	9.48	2.33
Ours			88.02	78.72	7.90	2.15
随机 无CP	8(10%)	72(90%)	87.93	78.67	8.24	2.08
Ours			88.75	79.88	7.66	1.83
			89.62	81.31	6.81	1.76

表 9. 在LA数据集上预训练策略的消融研究。random：随机初始化。w/o CP：从在没有复制粘贴的情况下使用有标签数据训练的预训练模型初始化。

BCP	nms	预训练	Dice↑	Jaccard↑	95HD↓	ASD↓
			47.62	36.61	29.02	11.46
✓			83.26	72.71	23.90	7.49
✓	✓		82.33	72.76	9.78	4.74
✓	✓	✓	87.59	78.67	1.90	0.67

表 10. 在 ACDC 数据集上使用 5% 有标签数据的消融实验， $\alpha = 0.5$ 所有实验。nms：对未标注数据的伪标签进行后处理。Pre-Train：使用在标注数据上进行 copy-paste 的预训练模型进行初始化。

随机掩码和接触掩码在缓解有标注数据与无标注数据之间的分布不匹配问题方面能力都较弱。

M 中零值区域的尺寸 我们在 LA 数据集上研究了零值区域大小的影响，如表 7 所示。对于掩码中的零值区域 $\beta H \times \beta W \times \beta L$ $\mathcal{M}$ ，我们设置 $\beta = \{\frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{5}{6}\}$ 。随着 β 减小，性能变差，这意味着小的复制粘贴前景在向/从背景传递共同语义方面的能力有限。当 $\beta = 2/3$ 时性能最佳，而当 $\beta = 5/6$ 时性能略有下降。

损失函数中的权重 我们将 $\alpha = 0.5$ 设为默认值。现在我们改变 $\alpha = \{0.5, 1.5, 2.5\}$ 以观察性能如何变化。表 8 显示，当 α 从 0.5 变化到 1.5 时，性能不敏感，但当 $\alpha = 2.5$ 时观察到明显的性能下降。

教师网络初始化策略在我们的默认设置中，教师网络由预训练模型初始化，该模型在有标注数据上以复制粘贴方式训练。我们研究了其他网络初始化策略：随机初始化和从预训练模型初始化，后者是在有标注数据上训练但未使用复制粘贴。比较结果如表9所示。与预

在原始有标注数据上训练相比，在预训练期间对有标注数据执行复制粘贴可以有效提升网络的泛化能力。

在5%有标签数据ACDC上的消融实验 在表 3中，我们的方法在仅使用5%有标签数据的ACDC数据集上取得了巨大提升。我们将方法拆分为三个组件，并研究哪个组件对该提升贡献最大。如表10所示，去掉我们的组件后（第一行），方法退化为普通的基于伪标签的自训练方法，这意味着有标签图像和无标签图像的分割分别由真值和伪标签监督。随后，BCP带来了显著的性能提升（Dice从47.62%提升至83.26%）。后处理（非极大值抑制）和更好的教师网络初始化提升了伪标签的质量，从而进一步提高了性能。

5. 结论

我们提出了用于半监督医学图像分割的双向复制粘贴（BCP）。我们以双向方式扩展了基于复制粘贴的方法，从而缩小了有标签数据和无标签数据之间的分布差距。在LA、NIH-Pancreas和ACDC数据集上的实验表明了所提出BCP的优越性，在仅使用5%有标签数据的ACDC数据集上Dice提升甚至超过21%。值得注意的是，与骨干网络相比，BCP并未引入新的参数或计算成本。**局限性。**我们没有专门设计用于增强局部属性学习的模块。尽管BCP优于所有对比方法，但对对比度极低的目标部位仍然难以很好地分割（例如，图4第二行左下部分缺失）。

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